Intrinsic Decomposition Embedded Spectral Unmixing for Satellite Hyperspectral Images With Endmembers From UAV Platform

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***Abstract*— Traditional spectral unmixing (SU) of satellite hyperspectral images (HSIs) faces two main challenges: one is that limited by the low resolution of satellite HSIs, it is difficult to guarantee the accuracy of endmember extraction due to severe spectral mixing; the other is that the spectral variability is unavoidable due to external factors such as atmospheric, illumination, and topographic variations, as well as internal factors such as physical changes of the features themselves. Unmanned aerial vehicle (UAV) HSIs of high spatial resolution can provide highly accurate reflectance curves from regions of interests (ROIs), and the intrinsic image decomposition (IID) technique can reduce the spectral variability caused by external factors. Based on this, a novel IID-embedded UAV-satellite SU model is proposed. On the one hand, the spectral variability is solved by an embedded IID framework in the inverse problem of SU. The proposed method replaces the input, i.e., the original HSI, with the reflectance component, which is independent of the spectral variability caused by external factors. On the other hand, a UAV spectral library constructed from the UAV HSI is introduced to guarantee the accuracy of the endmember. Thus, by IID embedded in the framework of UAV-satellite collaborative SU, the proposed method is able to address the aforementioned problems. Experimental validation is conducted using UAV HSI and three sets of satellite HSI from the Yellow River Delta (YRD) region. The results indicate that the proposed method can effectively improve the robustness and superiority of the unmixing results.**

***Index Terms*— Intrinsic image decomposition (IID), linear unmixing, satellite hyperspectral, spectral variability, unmanned aerial vehicle (UAV) hyperspectral.**

I. Introduction

In RECENT years, hyperspectral images (HSIs), with rich spectral information, have made significant contributions to various fields, including vegetation and agricultural monitoring [1], geological and mineral surveys [2], environmental monitoring, marine remote sensing [3], target detection [4], and field monitoring [5]. However, due to the poor spatial resolution of satellite HSI, the spectra of pixels often represent a combination of materials rather than a single material, which inevitably limits the further exploitation of hyperspectral data [6]. Therefore, spectral unmixing (SU) has been proposed and become increasingly important in HSI analysis and is now widely used in several applications [7].

The main purpose of hyperspectral SU is to obtain the categories of endmembers contained in the mixed image and to estimate the proportion of endmembers through endmember extraction and abundance estimation. Typical endmember extraction methods, including PPI [8], N-FINDR [9], vertex component analysis (VCA) [10], and simple growth algorithm (SGA) [11], can obtain the spectral information of the image, and the accuracy of endmember extraction often determines the quality of SU results. Although many endmember extraction methods have been proposed, there are still two major challenges: one is that the low resolution of satellite HSIs makes it difficult to guarantee the accuracy of endmember extraction; the other is that the complex imaging conditions and external variations lead to the degradation of the spectral information during the imaging process.

Due to external variations in the atmosphere, illumination, environmental conditions [12], and internal variability of the features themselves [12], [13], the phenomenon known as spectral variability [14] occurs. Fitting variability to this situation by modeling endmembers has received much attention from researchers. In [15], the perturbed linear mixing model (PLMM) was proposed to represent the spectral variability, where the variability is explained by the perturbation term applied to the endmembers, with the shortcoming that the model lacks physical meaning. The extended LMM (ELMM) [16] was proposed to model the spectral variability by scaling factors, and the generalized LMM (GLMM) [17] was later proposed by introducing an individual scaling factor for each band, but it is incomplete to consider only scaling factors. Augmented LMM (ALMM) [18], where the scaling factors are modeled by endmember dictionaries and an additional dictionary is proposed to model the rest of the spectral variables. The robust unmixing by variable splitting and augmented Lagrangian (RUSAL) [19] model assumes that the LMM is incomplete and considers the presence of spectral variability or anomalous pixels by imposing an additive term that can model the error factor. The modeling of spectral variability allows the

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modeling of uncertainty, but the physical implications of the model are different for different practical scenarios. Another general idea for spectral variability is to perform unmixing using spectral libraries. Methods based on spectral library unmixing consist of multiple endmember spectral mixture analysis (MESMA) [20], [21] and sparse SU [22], [23], [24]. These methods are usually conceptually simple and easy to explain, but the unmixing results depend heavily on the quality of the spectral library. The algorithms based on spectral library assume that the actual endmembers in the scene are consistent with the relevant spectra in the spectral library. However, due to the differences in imaging systems, the spectral information acquired based on spectral libraries has no spatial resolution. Furthermore, there are differences in imaging conditions, such as different imaging time, the actual endmembers of the image to be unmixed often do not match the corresponding spectral signals in the spectral libraries, which has an impact on the unmixing processing. Physics-based unmixing methods allow for a clear physical explanation, however, the unmixing results may not perform well dealing with the real complex scenes due to the limited modeling capabilities. On the other hand, data-driven deep learning methods benefit from the ability to model complex scenes and turn the approach into a "black box," but have the disadvantages of lacking a physical model to explain why a data-driven DNN can perform unmixing, as well as an over-parameterized structure that requires many training samples, resulting in low interpretability and poor robustness. Thus, the integration of physics-based and data-driven models for unmixing has received increasing attention [29]. The deep autoencoder (AE) network consists of an encoder and a decoder, with the encoder to seek a low-dimensional representation (e.g., the abundance) and the decoder to reconstruct the data with the corresponding basis (e.g., the endmember), which also guarantees a physical interpretation. The AE framework has been proven to be successful for SU [30], [31], [32], [33], [34], [35], [36], [37]. Qu and Qi [30] describes an untied denoising autoencoder with sparsity (uDAS) that introduces a non-negative sparsity and denoising coding module to obtain better unmixing performance. Convolutional neural networks (CNNs) have also been used for encoder design [35], [36]. As the 2-D convolutional kernel focuses mainly on spatial structure, the 3-D CNN encoder [36] is used to combine local spectral features. Two cascaded encoders are trained end-to-end in [37] to effectively improve the unmixing performance. For spectral variability, in addition to modeling the endmembers to fit the variability, intrinsic image decomposition (IID) [25], [26], [27], [28] can recover from a single HSI the reflectance properties related to the material's own reflectance characteristics and directly reduce the spectral variability caused by atmosphere, ambient lighting, and topographic relief during the imaging process.

To solve the problem mentioned above, a novel IID-embedded unmanned aerial vehicle (UAV)-satellite collaborative SU model is carried out, named IID embedded SU (IIDSU). As the name suggests, by embedding the IID model into the SU model to reduce spectral variability, and by introducing high-resolution UAV HSI data to ensure the reliability of the target reflectance curve, IIDSU can achieve more accurate and robust unmixing results. Specifically, the pixels from UAV HSI (centimeter level spatial resolution) can be considered as pure endmembers compared to those from satellite HSI (meter level spatial resolution). The target land covers of interest are selected from the UAV HSI and their reflectance curves are extracted to construct a prior UAV spectral library [38], [39], which is considered as a standard endmember spectral library containing target species of interest at similar phenological stages since the main features of interest are vegetation types. Besides, spectra acquired from the UAV can average intraclass difference compared to that from ground point spectrometer, thus ensuring a more physical endmember input. Furthermore, the UAV spectral library effectively avoids the mismatch caused by using typical spectral library [20], [21], [22], [23], [24], which is a breakthrough in ensuring the accuracy of endmember extraction.

In this article, different from the traditional SU methods, a more accurate UAV spectral library is selected as a reference in consideration of the difficulty in guaranteeing the accuracy of endmember extraction. Besides, the influence of spectral variability can be effectively reduced by the embedded IID model. From the perspective of the physical model, the proposed IIDSU model can ensure the effectiveness and robustness of SU. In addition, the framework of the proposed model is universal and can be extended to other unmixing models, reflecting its superiority over other methods.

More specifically, the innovative points are as follows:

1) An IID-embedded hyperspectral unmixing framework was proposed to reduce the spectral variability caused by changes in the imaging process and provide robust unmixing results in the physical sense. Specifically, we replace the original hyperspectral with the intrinsic reflectance hyperspectral as the input, making it more accurate to process the SU and improving the robustness and effectiveness.
2) A UAV-satellite collaborative SU model was proposed to provide a new direction to solve the difficulty in ensuring the accuracy of endmember extraction in traditional LMM due to the limited spatial resolution. Concretely, by considering the high resolution of UAV HSI, the reflectance curves of land covers acquired at similar phenological stages can be treated as reference spectra. Besides, by achieving the dimensional alignment of UAV HSI and satellite HSI, the difficulty in guaranteeing the accuracy of endmember extraction in traditional SU can be effectively overcome.

The rest of this article is organized as follows. Section II introduces the related work. Section III describes the proposed model and the corresponding optimization algorithm. Section IV presents experimental results on three different datasets and discusses the qualitative and quantitative analysis. Finally, Section V concludes.

II. Related Work

In this section, the LMM model and classical SU methods were introduced first, the motivation of these unmixing models was presented and analyzed in detail and their advantages and

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disadvantages were highlighted, and finally, the fundamental imaging theory of IID was introduced.

A. Linear Mixing Model

The LMM [40] is a simple but effective model. It assumes that the reflecting surface is a regular mixture of ground and plane, and does not consider the interaction between the light and the material, the formation of mixed image endmembers is mainly due to the low resolution of the imaging spectrometer.

In the LMM model, for each pixel in the spectral vector $y_n, n = 1, \ldots, N, y_n \in \mathbb{R}^{L\times 1}$ in the satellite HSI matrix $Y \in \mathbb{R}^{L\times N}$ of L-band, the LMM can be expressed as

$$y_n = \mathbf{E}_0 a_n + n_n, \quad \text{s.t. } \mathbf{1}^T a_n = 1 \text{ and } a_n \geq 0 \tag{1}$$

where $\mathbf{E}_0 = [e_{0,1}, \ldots, e_{0,P}]$ is an endmember matrix of size $L \times P$, P represents the number of endmembers. a_n is the abundance vector.

The above expressions can be expressed in matrix form as:

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N} \tag{2}$$

where $\mathbf{E} \in \mathbb{R}^{L\times P}$ represents the endmember matrix, $\mathbf{A} \in \mathbb{R}^{P\times N}$ represents the abundance matrix, and $\mathbf{N} \in \mathbb{R}^{L\times N}$ is the observed noise matrix. There are two constraints on the abundance matrix as follows: 1) the abundance non-negativity constraint (ANC), namely $\mathbf{A}_{i,j} \geq 0$, and 2) the abundance sum-to-one constraint (ASC), namely $\mathbf{1}_P^T a_n = 1 (\mathbf{1}_P = [1, 1, \ldots, 1]^T \in \mathbb{R}^P)$.

In the LMM model, the inverse solution to the abundance matrix can be expressed as the following optimization problem:

$$\min_{a_i} \|\mathbf{E}a_i - y_i\|_F^2, \quad i = 1, \ldots, N \tag{3}$$

then $\bar{a}_i = (\mathbf{E}^T\mathbf{E})^{-1}\mathbf{E}^T\bar{y}_i$ can be obtained, obviously, the optimization problem is not constrained to abundance. Considering the ASC, the matrix can be expressed in the augmented form as $\bar{y}_i = \begin{bmatrix} y_i \\ \mathbf{1} \end{bmatrix}^T$, $\bar{\mathbf{E}} = \begin{bmatrix} \mathbf{E} \\ \mathbf{1}^T \end{bmatrix}^T$. The least squares analytic solution under the ASC can be obtained as $a_i = (\bar{\mathbf{E}}^T\bar{\mathbf{E}})^{-1}\bar{\mathbf{E}}^T\bar{y}_i$.

By combining ASC and ANC, the fully constrained linear SU (FCLSU) model can be obtained. The FCLSU method is theoretically easy to understand and achieve, yet the abundance obtained by FCLSU does not fully correspond to the surface material distribution due to the complexity of the imaging environment and other factors. FCLSU is generally used as the initialization of abundance estimation.

B. SU Based on Spectral Variability

In real-world imaging, SV often occurs due to significant variations in changing imaging conditions. In many cases, a better unmixing result can be achieved by building a model for the spectral variability.

1) Sparse Unmixing: For HSI $\mathbf{Y} \in \mathbb{R}^{D\times N}$, given a complete and over-redundant spectral library $\mathbf{E} \in \mathbb{R}^{D\times K}$, where D and K are the number of spectral bands and feature spectra contained in the spectral library, respectively. Because P is much smaller than K, the corresponding abundance has a significant sparse property.

The sparse unmixing problem can be formulated as the process of selecting a subset of endmembers in the over-redundant spectral library that minimizes the reconstruction error, thus the ANC on the abundance matrix can be transformed into a sparsity constraint-based regression problem, which has the matrix representation shown below

$$\min_A \left\{ \frac{1}{2}\|\mathbf{Y} - \mathbf{E}\mathbf{A}\|_F^2 + \lambda\|A\|_1 \right\}, \quad \text{s.t. } A \geq 0 \tag{4}$$

where $\|\cdot\|_1$ denotes the sparsity-inducing $L1$ norm of the vectorized matrix, $\lambda \geq 0$ is a penalty parameter to balance the importance of different terms in (4). Here, the ASC is relaxed because of its incompatibility with the $L1$ norm. However, the results of the Sparse Unmixing model might be sensitive to the selection of the regularization coefficient λ and might not be as effective as the method without sparse regularization term.

2) ELMM: The PLMM represents the spectral variability mainly by applying an additive factor, however, since the additive factor is not sufficiently complete to fit the spectral variability in real datasets, the ELMM appears by applying an additional scale factor. The ELMM focuses on modeling the spectral variability of the target endmember by adding scaling factors to adjust the endmember. The model can be expressed as

$$y_k = \sum_{p=1}^{P} a_{kp}\psi_{kp}\mathbf{E}_p + n_k = \mathbf{E}_0\psi_k a_k + n_k \tag{5}$$

where $\mathbf{E}_0 \in \mathbb{R}^{D\times P}$ represents the spectral variability, and $\psi_i \in \mathbb{R}^{P\times P}$ is a diagonal array.

The ELMM focuses mainly on modeling spectral variability by scaling factors, and its shortcoming is that the modeling is not complete when faced with more complex spectral variability. Specifically, the ELMM model lacks the flexibility to represent more complex spectral variability despite its physical motivation.

3) RUSAL: In addition to considering the additivity factor and scale factor applied to fit the spectral variability of the endmembers, the real satellite HSI data are often subject to sparse outliers or other unknown effects (impulse noise, horizontal, vertical line streaks, etc.), where it is desired to make the model more robust. Error modeling is introduced by considering spectrally smoothed residuals as follows:

$$\begin{aligned} y_k &= \mathbf{E}a_k + \mathbf{F}^T b_k + n_k \\ \phi_k(b_k) &= \mathbf{F}^T b_k \end{aligned} \tag{6}$$

where $\mathbf{F} \in \mathbb{R}^{C\times L}$ denotes the first C rows after the DCT transformation, b_k is the coefficient of the DCT transformation, the product of two matrices accounts for spectral variability and other unmodelled effects. ϕ_k is the smoothing spectral function, where the smoothness is guaranteed by applying sparsity constraints. RUSAL is equivalent to the LMM when

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b_k is 0. However, the RUSAL method has a more limited modeling capability. The resulting optimization problem is non-convex. The first matrix $\mathbf{F} \in \mathbb{R}^{C\times L}$ corresponded to the first columns of the discrete cosine transform, forcing the additive terms to be spectrally smooth. The second matrix b_k was defined for the pixel-dependent coefficients, which were forced to be spatially sparse and were estimated by solving a convex optimization problem.

C. Intrinsic Image Decomposition

From the remote sensing imaging model, IID can be expressed specifically as follows: the HSI $\mathbf{Y}$ can be decomposed into a reflectance component $\mathbf{R}$ that characterizes the properties of the material itself and a shading component $\mathbf{S}$ that characterizes the interaction of ambient light with the surface normal of the material.

For each pixel in the spectral vector $\boldsymbol{y}_n \in \mathbb{R}^{L\times 1}$ in a satellite HSI $\mathbf{Y} \in \mathbb{R}^{L\times N}$, the IID can be expressed as follows:

$$y_n = s_n \odot r_n \tag{7}$$

where $s_n \in \mathbb{R}^{L\times 1}$, $r_n \in \mathbb{R}^{L\times 1}$, the mathematical symbol $\odot$ denotes the Schur-Hadamard product.

The above equation can be expressed in matrix form as

$$\mathbf{Y} = \mathbf{S} \odot \mathbf{R} \tag{8}$$

where $\mathbf{S} \in \mathbb{R}^{L\times N}$, $\mathbf{R} \in \mathbb{R}^{L\times N}$. To make the ill-posed problem solvable, a constraint matrix about $\mathbf{R}$ was further defined where $\mathbf{S}$ was initialized as an all-1 matrix here.

The existing common reflectance prior is a smoothing prior

$$\mathbf{R}_i = \sum_{j\in M_i} \omega_{ij}\mathbf{R}_j \tag{9}$$

where $\mathbf{R}_i$ and $\mathbf{R}_j$ are the reflectance of the pixel i and the pixel j, respectively, M_i is the local window of the pixel i, and ω_{ij} measures the spectral similarity between the pixels i and the pixel j. The disadvantage of the smoothing prior is that the assumptions are not supported by the boundaries. To solve this problem, the reflectance prior based on ℓ_1-map encoding is extended to make the reflectance more robust and representative using sparse constraints, and the reflectance prior matrix W is established based on the results of ℓ_1-map encoding

$$W_{ij} = \begin{cases} \alpha_j^i, & i > j \\ \alpha_{j-1}^i, & i < j \end{cases} \tag{10}$$

thus, the reflectance priori constraint can be expressed as

$$\Psi(\mathbf{R}) = \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2. \tag{11}$$

III. Proposed Approach

A. Motivation

HSI is limited by the low spatial resolution, resulting in a small area being characterized as a mixed pixel, and areas within the same mixed pixel are assumed to have a similar shading component due to the presence of similar topography and illumination conditions. Considering that the shading component characterizes the changes in the imaging process due to lighting conditions and topographic variations, while the intrinsic reflectance component characterizes the features that are only related to the material properties, the IID can effectively reduce the influence of spectral variability by reducing the shading component and recovering the intrinsic reflectance component. Spectral reflectance can be effectively improved by improving the homogeneity of spectral reflectance within a class. The effective reduction of spectral variability can greatly enhance the subsequent improvement in unmixing tasks, which led us to further improve the LMM model by adapting the original HSI to an intrinsic HSI that is not subject to changes in external imaging conditions.

The adjustment modeling can be formulated as

$$\mathbf{R} = \mathbf{E}\mathbf{A} + \mathbf{N}. \tag{12}$$

In conclusion, the model can be formulated as follows:

$$\mathbf{Y} = \mathbf{S} \odot (\mathbf{E} \cdot \mathbf{A}). \tag{13}$$

It is a pioneering model for an IID-embedded LMM, which in a physical sense will greatly improve the robustness of unmixing in complex scenarios with real data.

To better understand the proposed framework, it is included in the proposed model as an example of Sparse SU, the details of which are solved and derived below.

B. Dimensional Alignment: From UAV HSIs to Satellite HSIs

The endmember matrix used for unmixing is constructed based on the UAV HSI. Considering the differences between UAV and satellite (sensors, spectral ranges, and spectral resolution) are inevitable. Therefore, dimensional alignment between UAV HSI and satellite HSI needs to be introduced.

The UAV spectral library $\mathbf{E}_{\text{uav}}$ acquired from the UAV-HSI can be considered as reference spectral reflectance curves obtained by selecting a set of pixels for each land cover of interest in the UAV HSI. Specifically, the UAV HSI data are first normalized and then a set of reflectance curves is obtained from several pixels uniformly selected based on a priori knowledge of the land cover. Under the premise that the reflectance curves satisfy the objective physical distribution, the maximum, mean plus variance, mean, mean minus variance, and minimum value of the spectral curves for each feature type are extracted so that a total of five characteristic spectral curves can be obtained for each land cover of interest. In this way, the accuracy of the spectral curves of the land covers can be guaranteed. The $\mathbf{E}_{\text{uav}}$ can be spectrally down-sampling by the spectral response function (SRF) to achieve dimensional alignment with the satellite HSI

$$\overline{\mathbf{E}}_{\text{uav}} = \mathbf{F}\mathbf{E}_{\text{uav}} \tag{14}$$

where $\mathbf{F} \in \mathbb{R}^{D_{\text{sat}}\times D_{\text{uav}}}$ describes the SRF, D_{sat} and D_{uav} characterize the spectral dimension of SATHSI and UAV HSI, respectively, $\mathbf{E}_{\text{uav}} \in \mathbb{R}^{D_{\text{uav}}\times P}$ represents the original dimension of the UAV spectral library, $\mathrm{E}_{\text{uav}} = [\mathrm{E}_1, \mathrm{E}_2, \ldots, \mathrm{E}_P]$, $\overline{\mathbf{E}}_{\text{uav}} \in \mathbb{R}^{D_{\text{sat}}\times P}$ represents the UAV spectral library that has realized dimensional alignment, $\overline{\mathbf{E}}_{\text{uav}} = \left[\overline{\mathbf{E}}_1, \overline{\mathbf{E}}_2, \ldots, \overline{\mathbf{E}}_P\right]$.

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In summarizing, the proposed IID-embedded UAV-to-satellite SU model can be formulated as

$$\mathbf{Y} = \mathbf{S} \odot \left(\overline{\mathbf{E}}_{\text{uav}} \cdot \mathbf{A}\right). \tag{15}$$

In the following, we will replace $\overline{\mathbf{E}}_{\text{uav}}$ with $\mathbf{E}$. It is important to be clear that the IID-embedded framework provides robust unmixing by effectively reducing the effect of spectral variability, while the use of a UAV-satellite collaborative model, with endmembers from UAV avoids the problem of inaccurate endmember extraction. Considering that the spectra of the satellite HSI themselves may achieve smaller unmixing errors for unmixing due to the high data matching, the use of UAV spectra provides a further way of unmixing for similar feature types of interest and avoids the prior information required for satellite HSI.

In fact, a given spectral signature is usually composed of a limited number of materials in a hyperspectral scene. Therefore, the sparsity of the abundance matrix needs to be considered.

We then applied $\|\mathbf{A}\|_{1,1} \equiv \sum_{n=1}^{N} \|a_n\|_1$ to approximately estimate the sparsity-prompting term, which can be expressed with the penalty parameter γ as

$$\Upsilon(\mathbf{A}) = \gamma \|\mathbf{A}\|_{1,1}. \tag{16}$$

C. Optimization Algorithm

The priors introduced in Sections II-B and III-C, the following problem is defined to perform the optimization solution

$$U = \frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \Psi(\mathbf{R}) + \Upsilon(\mathbf{A}) \quad \text{s.t. } \mathbf{A} \geq 0 \tag{17}$$

the terms $\Psi(\mathbf{R})$ and $\Upsilon(\mathbf{A})$ represent the reflectance constraints and abundance constraints, respectively. Specifically, where the intent is to estimate the variables $\mathbf{S}$, $\mathbf{R}$, and $\mathbf{A}$, while $\mathbf{E}$ is given, the basic objective function of IIDSU also can be expressed as

$$U = \frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2 + \gamma\|\mathbf{A}\|_{1,1} \tag{18}$$

$$\text{s.t. } \mathbf{A} \geq 0$$

$$\arg\min_{\mathbf{S},\,\mathbf{R},\,\mathbf{A}} U(\mathbf{S},\mathbf{R},\mathbf{A}). \tag{19}$$

In (20), the first term aims at keeping the residual error between the original HSI $\mathbf{Y}$ and the reconstructed HSI $\mathbf{S} \odot \mathbf{R}$ as small as possible. Similarly, the second term is to minimize the residual error between the reflectance component $\mathbf{R}$ and the reconstructed HSI during the process of unmixing.

Currently, there is no algorithm specifically designed to solve the model proposed in this article based on the available reference materials.

Here, an alternating direction method of multipliers (ADMMs) algorithm is introduced to efficiently and quickly optimize the solution of the above model. The above-mentioned objective function can be reformulated into an equivalent form by introducing auxiliary parameters $\mathbf{R} = \mathbf{J}$ and $\mathbf{A} = \mathbf{K}$. In particular, the shading component is generally initialized by $\mathbf{S}\mathbf{1}_{N\times N} \in \mathbf{1}_{L\times N}$, the reflectance component is normally initiated to $\mathbf{Y}$, and the abundance matrix is generally initialized by FCLSU

$$\min_{M} \frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2 + \alpha\|\mathbf{K}\|_1 + \ell_R^+(\mathbf{K}) \quad \text{s.t. } \mathbf{R} = \mathbf{J}, \quad \mathbf{A} = \mathbf{K}, \ \mathbf{1}^T\mathbf{A} = \mathbf{1} \tag{20}$$

where $M = \{\mathbf{S}, \mathbf{R}, \mathbf{A}, \mathbf{J}, \mathbf{K}\}$, $\ell_R^+(\mathbf{K})$ means $\mathbf{K} \geq \mathbf{0}$. The augmented Lagrangian function can be expressed in the following way:

$$\begin{aligned} &L(M, \Lambda_1, \Lambda_2) \\ &= \frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2 \\ &\quad + \alpha\|\mathbf{K}\|_1 + l_R^+(\mathbf{K}) + \Lambda_1^T(\mathbf{J} - \mathbf{R}) + \Lambda_2^T(\mathbf{K} - \mathbf{A}) \\ &\quad + \frac{u}{2}\|\mathbf{J} - \mathbf{R}\|_F^2 + \frac{u}{2}\|\mathbf{K} - \ \mathbf{A}\|_F^2 \\ &\quad \text{s.t. } \mathbf{1}^T\mathbf{A} = \mathbf{1} \end{aligned} \tag{21}$$

where Λ_1, Λ_2, and u are two Lagrange multipliers and a positive penalty parameter, respectively.

For each of the variables involved, the optimal solution follows.

Optimization with respect to $\mathbf{R}$

$$\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{Y} - \mathbf{S} \odot \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_F^2 + \Lambda_1^T(\mathbf{J} - \mathbf{R}) + \frac{u}{2}\|\mathbf{J} - \mathbf{R}\|_F^2 \tag{22}$$

here a track is used by introducing a new variable $\mathbf{T}$, which satisfies $\mathbf{S} \odot \mathbf{T} = \mathbf{Y}$, so that the following equation can be obtained:

$$\min_{\mathbf{R}} \frac{1}{2}\|\mathbf{T} - \mathbf{R}\|_F^2 + \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \frac{1}{2\sigma}\left\|\mathbf{R} - \mathbf{R}\mathbf{W}^T\right\|_{\text{F}}^2 + \Lambda_1^T(\mathbf{J} - \mathbf{R}) + \frac{u}{2}\|\mathbf{J} - \mathbf{R}\|_F^2 \tag{23}$$

which provides a closed solution

$$\mathbf{R} = \left(\mathbf{T} + \lambda\mathbf{E}\mathbf{A} + u\mathbf{J} + \Lambda_1\right) \times (\mathbf{N} + \mathbf{O})^{-1} \tag{24}$$

where

$$\mathbf{N} = u + \lambda + 1 \tag{25}$$

$$\mathbf{O} = \frac{1}{\sigma}\left(1 - 2\mathbf{W}^T + \mathbf{W}^T\mathbf{W}\right). \tag{26}$$

Optimization with respect to $\mathbf{S}$:

Considering $\mathbf{R}_{k+1} = \left(\mathbf{T}_k + \lambda\mathbf{E}\mathbf{A} + u\mathbf{J} + \Lambda_1\right) \times (\mathbf{N} + \mathbf{O})^{-1}$, according to $\mathbf{S}_{k+1} \odot \mathbf{R}_{k+1} = \mathbf{Y}$, $\mathbf{S}_{k+1}$, and $\mathbf{T}_{k+1}$ both can be subsequently obtained.

Optimization with respect to $\mathbf{A}$

$$\min_{\mathbf{A}} \frac{\lambda}{2}\|\mathbf{R} - \mathbf{E}\mathbf{A}\|_F^2 + \Lambda_2^T(\mathbf{K} - \mathbf{A}) + \frac{u}{2}\|\mathbf{J} - \mathbf{R}\|_F^2 + \frac{u}{2}\|\mathbf{K} - \mathbf{A}\|_F^2 \tag{27}$$

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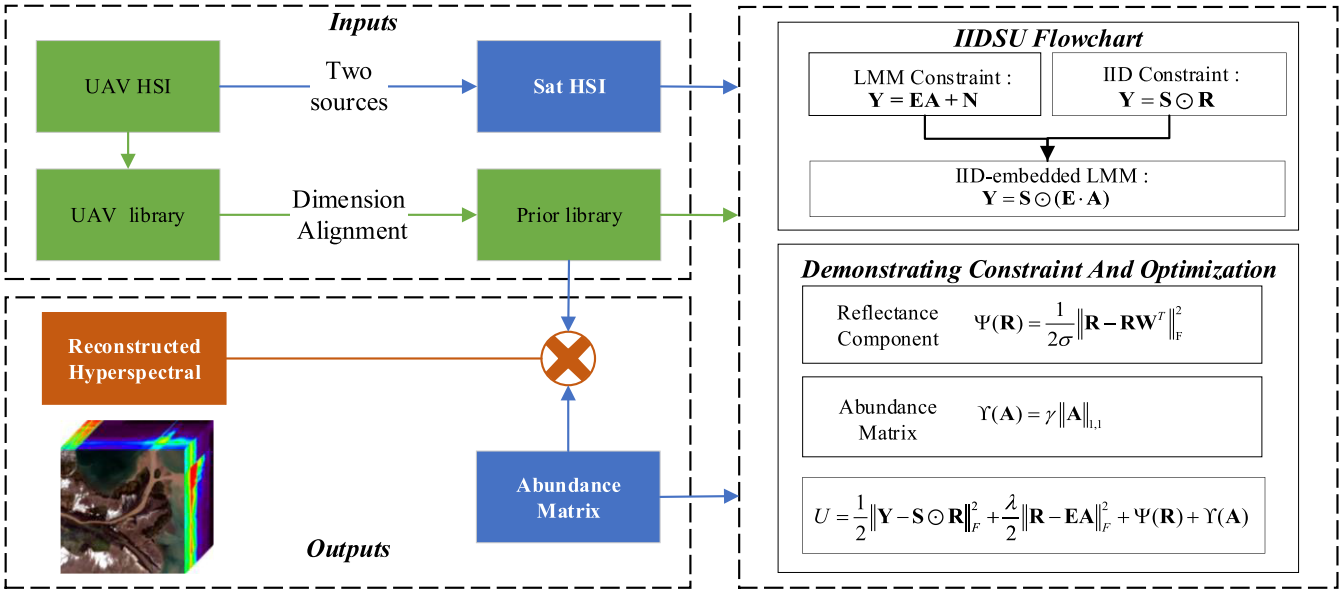


Fig. 1. Framework of the proposed IIDSU algorithm. The inputs include the satellite HSI to be unmixed and the UAV spectral library. IIDSU model shows the matrix form of the proposed algorithm. Sparse SU as an example is shown next to demonstrate the IIDSU model. Then the outputs can be obtained as the abundance matrix and the reconstructed HSI.

Algorithm 1

1: Give the input image $\mathbf{Y}$, a UAV spectral library $\mathbf{E}$ that has realized dimensional alignment.
2: Initialize the shade component $\mathbf{S1}_{N\times N} \in \mathbf{1}_{L\times N}$, the abundance matrix $\mathbf{A}$ by FCLSU, the reflectance component $\mathbf{R} = \mathbf{Y}$, $\mathbf{W1}_{N\times N} = \mathbf{1}_{N\times N}$, $\mathbf{J} = \mathbf{0}$, $\mathbf{K} = \mathbf{0}$, $\Lambda_1 = \Lambda_2 = \mathbf{0}$.
3: **repeat:**
4: Update the variable $\mathbf{R}$ by (21);
5: Update the variable $\mathbf{A}$ by (22);
6: Update the variable $\mathbf{J}$ by (23);
7: Update the variable $\mathbf{K}$ by (24);
8: Update the variable Λ_1 and Λ_2 by (25);
9: **until:** Convergence

whose can be well solved by

$$\mathbf{A} = \left(\lambda \mathbf{E}^T\mathbf{R} + u\mathbf{K} + \Lambda_2\right) \times \left(\lambda \mathbf{E}^T\mathbf{E} + u\mathbf{I}_{P\times P}\right)^{-1}. \quad (28)$$

Optimization with respect to $\mathbf{J}$

$$\min_{\mathbf{J}} \Lambda_1^T(\mathbf{J} - \mathbf{R}) + \frac{u}{2}\|\mathbf{J} - \mathbf{R}\|_F^2 \quad (29)$$

which has an analytical resolution

$$\mathbf{J} = u\mathbf{R} - \Lambda_1/u. \quad (30)$$

Optimization with respect to $\mathbf{K}$

$$\min_{\mathbf{K}} \alpha\|\mathbf{K}\|_1 + \Lambda_2^T(\mathbf{K} - \mathbf{A}) + \frac{u}{2}\|\mathbf{K} - \mathbf{A}\|_F^2 \quad (31)$$

which can be derived by a soft threshold operator [41]

$$\mathbf{K} = \text{sign}(\mathbf{A} - \Lambda_2/u) \odot \max\{\mathbf{0}, |\mathbf{A} - \Lambda_2/u| - \alpha/u\}. \quad (32)$$

Optimization with respect to Λ_1, Λ_2.

The Lagrange multipliers Λ_1 and Λ_2 can be updated as following:

$$\begin{aligned} \Lambda_1 &\leftarrow \Lambda_1 + u(\mathbf{J} - \mathbf{R}) \\ \Lambda_2 &\leftarrow \Lambda_2 + u(\mathbf{K} - \mathbf{A}). \end{aligned} \quad (33)$$

Fig. 1 shows the flowchart of the proposed IIDSU algorithm, where the substitution into the IIDSU model is the Sparse SU algorithm. Other IIDSU models for the unmixing algorithm are conducted in the same way.

IV. Experimental Results and Discussion

A. Data Description

To validate the performance of the proposed unmixing algorithm in processing real HSIs, three HSI datasets containing the same geographical location, i.e., the YRD_Orbit HSI, the YRD_ZY1-02D image, and the YRD_GF5 image, as well as the UAV data, are used.

1) UAV-HSI: The UAV HSI was acquired in the Yellow River Delta (YRD) National Nature Reserve in China on 23 September, 2020. It contains 1623 × 1903 pixels and has 270 spectral bands with a spectral range from 0.4 to 1.0 μm. The spatial resolution is 0.12 m per pixel, and the spectral resolution is 6 nm. The UAV platform is the DJI Matrice 600 Pro. Fig. 2(a) shows the false color image. The reflectance curves of different land covers, including Spartina alterniflora, Suaeda salsa, Tamarisk, Phragmites, water and mudflat, are extracted from the high-resolution UAV data, as shown in Fig. 2(b). The spatial coverage relationship between UAV-HSI and satellite-HSI has been shown graphically in Fig. 2(c), the ground distribution of the UAV data has been supplemented in Fig. 2(d), which has been chosen to be shown here in a classification map. The labels for the six feature types of interest (Spartina alterniflora, Suaeda salsa, Tamarisk, Phragmites, water and mudflat) are shown in Fig. 2(e), and

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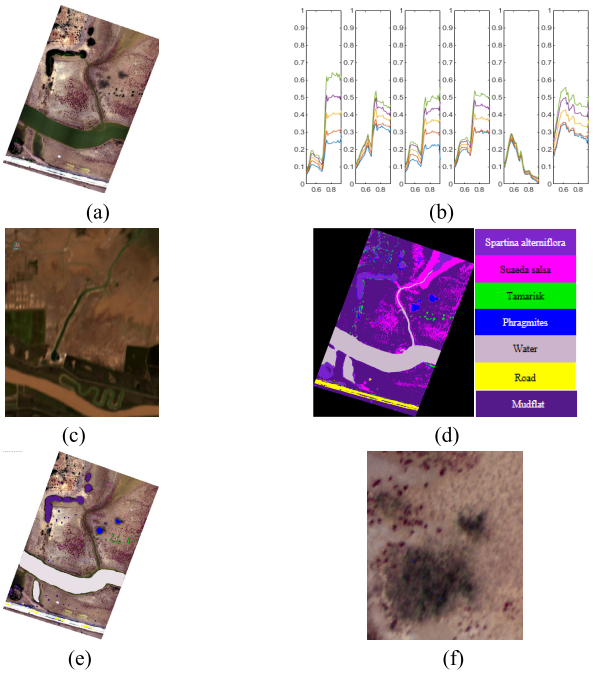


Fig. 2. UAV HSI data. (a) False color map. (b) Prior spectral library. (c) Spatial extent of the UAV. (d) Classification map. (e) Prior spectral library. (f) Prior spectral library of Phragmites.

TABLE I
INFORMATION FOR THE UAV DATA AND THE THREE SAT DATASETS

Datasets	Acquired Time	GSD/m	Fusion or not	Fusion GSD/m	Spectral Bands of 0.4 to 1.0 um
UAV	2020/9/23	0.12	×	/	270
SAT-Orbit	2020/9/30	10	×	/	32
SAT-ZY102D	2020/9/26	30	√	8	70
SAT-GF5	2019/5/31	30	√	8	140

Fig. 2(f) shows selected pixels to obtain a priori spectral libraries for Phragmites as an example.

It is worth noting that the spectral difference between the selected vegetation categories is quite small, which poses a new challenge for unmixing and provides a new direction for further research.

2) SAT-HSI:

a) YRD_Orbit: This Orbita Hyperspectral (OHS) data with 2086 × 2197 pixels was acquired at the YRD National Nature Reserve in China on 30 September, 2020. 32 spectral bands with a spectral coverage ranging from 0.4 to 1.0 μm were captured. The spatial resolution is 10 m. Fig. 3(a) shows false color image of this dataset.

b) YRD_ZY102D: The original 2603 × 2739 pixels image was acquired by the Advanced Hyper Spectral Imager (AHSI) on 26 September 2020. The AHSI sensor operates in the 0.4–2.5 μm wavelength range, contains 166 bands, including 76 bands in the VNIR range as well as 90 bands in the SWIR range. Fig. 3(d) presents false color image of this dataset.

c) YRD_GF5: This HSI contains 330 spectral bands with wavelengths ranging from 0.4 to 2.5 μm and was captured by the GF-5 satellite on 31 May, 2019. The size of this HSI is 2606 × 2745 pixels, with a spatial resolution of 30 m per pixel. Fig. 3(g) shows false color image of this dataset.

It should be noted that to reduce the difference in spatial resolution between the UAV HSI and the satellite HSIs, the latter two datasets were fused with the multispectral images (MSIs) in similar phenological stages to achieve a spatial resolution of 8 m [42].

TABLE II
RESULTS OF THE THREE DATASETS OF SUBBLOCK1

Subblock1	Methods	RMSE		PSNR	
		Ori	IID	Ori	IID
YRD_Orbit	FCLSU	0.0568	*0.0466*	20.3623	*20.66*
	Sparse	0.0564	*0.0400*	20.4057	*20.98*
	ELMM	0.0548	*0.0453*	20.6561	*20.72*
	RUSAL	0.0568	***0.0462***	20.3912	***21.22***
	uDAS	0.1480		20.43	
	Ideal	0		∞	
		Ori	IIDSU	Ori	IIDSU
YRD_ZY102D	FCLSU	0.0963	*0.0561*	16.7828	*20.4716*
	Sparse	0.0682	***0.0487***	16.8345	*20.4900*
	ELMM	0.0910	*0.0557*	17.1818	*20.5420*
	RUSAL	0.0958	*0.0546*	20.7147	***21.6602***
	uDAS	0.1951		12.75	
	Ideal	0		∞	
		Ori	IIDSU	Ori	IIDSU
YRD_GF5	FCLSU	0.1913	*0.1327*	12.8940	*15.6741*
	Sparse	0.1899	*0.1321*	12.9403	*15.7200*
	ELMM	0.1774	*0.1268*	13.6433	*16.1045*
	RUSAL	0.1489	***0.1032***	15.7244	***18.2513***
	uDAS	0.2192		16.18	
	Ideal	0		∞	

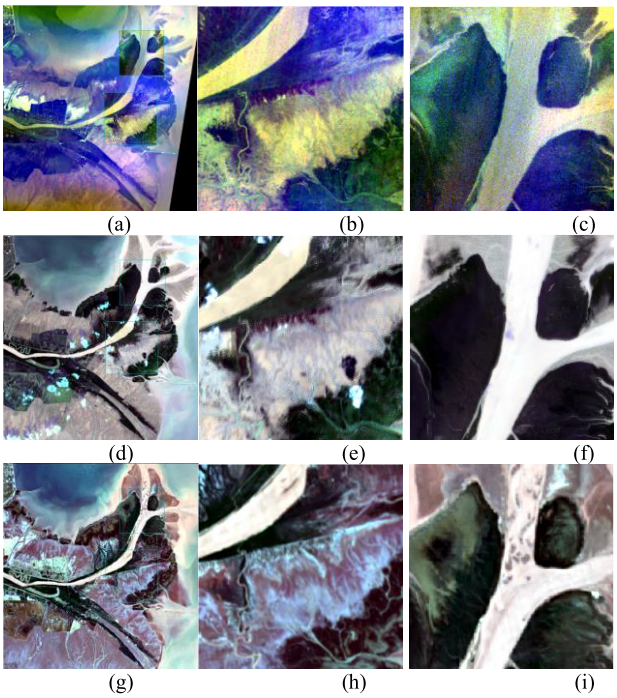


Fig. 3. False color images for the three satellite datasets. (a) YRD_Orbit dataset, (b) subblock1 of the YRD_Orbit dateset, (c) subblock2 of the YRD_Orbit dataset, (d) YRD_ZY102D dataset, (e) subblock1 of the YRD_ZY102D dataset, (f) subblock2 of the YRD_ZY102D dataset, (g) YRD_GF5 dataset, (h) subblock1 of the YRD_GF5 dataset, and (i) subblock2 of the YRD_GF5 dataset.

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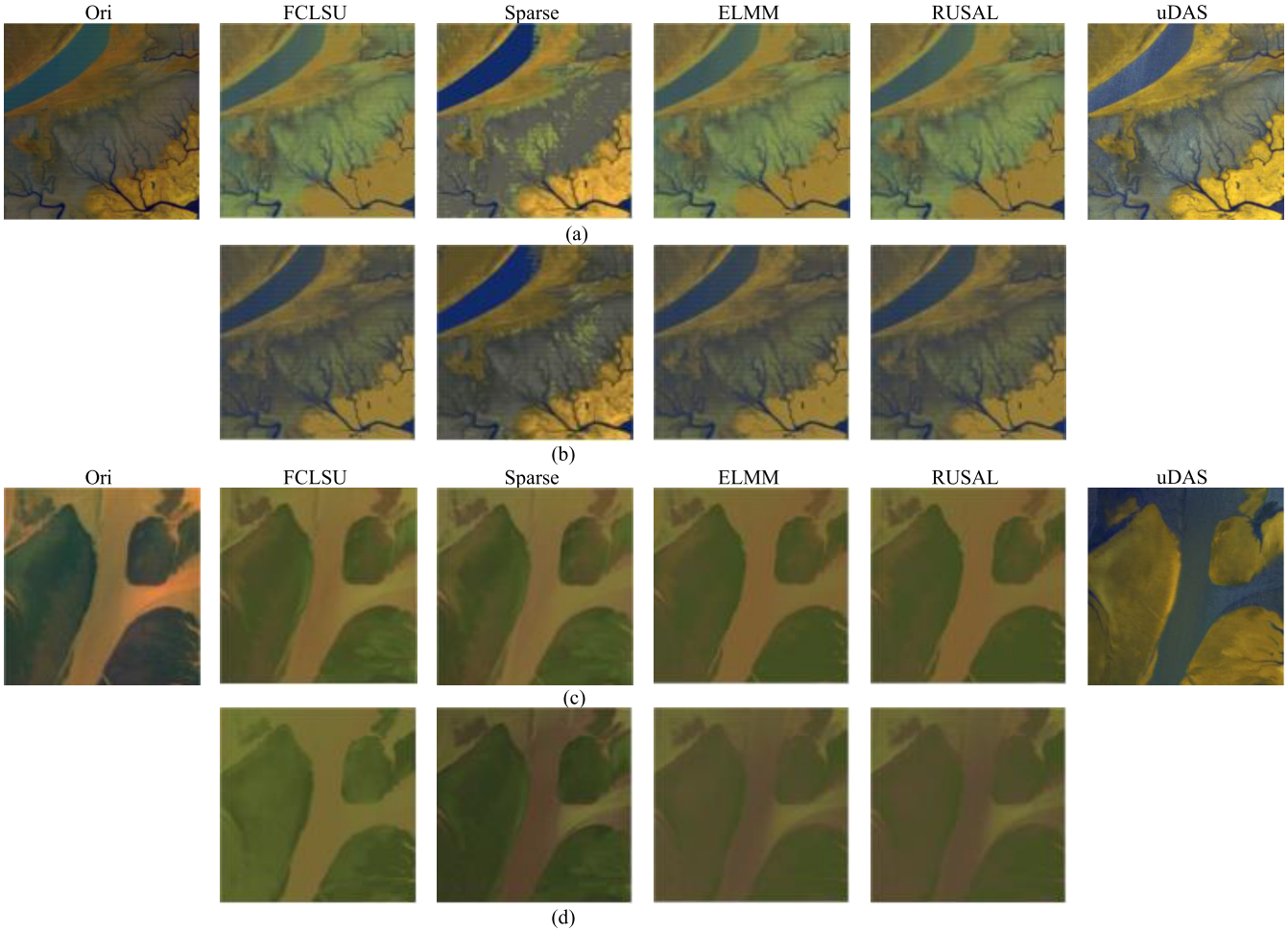


Fig. 4. Reconstructed results of unmixing (YRD_Orbit image) obtained by the original model and the proposed IIDSU model using FCLSU, Sparse SU, ELMM, RUSAL, and uDAS methods for two subblocks. (a) Subblock1-Ori. (b) Subblock1-IIDSU. (c) Subblock2-Ori. (d) Subblock2-IIDSU.

To verify the robustness of the method, two subblocks of the same region were selected in each of the three satellite HSI datasets above, as shown in Fig. 3. The YRD_ZY1-02D HSI needs to be masked first due to the existence of thick clouds. Fig. 3(b) and (c) show two subblocks of the YRD_Orbit dataset with sizes of 560 × 560 and 480 × 480, respectively. Fig. 3(e) and (f) show two subblocks of the YRD_ZY1-02D dataset with sizes of 700 × 700 and 600 × 600, respectively. Fig. 3(h) and (i) show two subblocks of the YRD_GF5 dataset with sizes of 700 × 700 and 600 × 600, respectively. Table I provides the detailed information of the UAV data and the three satellite datasets.

B. Evaluation Criteria

Since the abundance of information in the real HSIs is unknown, conclusions were drawn by analyzing the performance of the algorithm, assessed by the reconstruction overall root mean square error (RMSE_Y) and the peak signal-to-noise ratio (PSNR).

C. Experimental Results and Analysis

Typical algorithms based on spectral variability are incorporated into the proposed IIDSU framework through the processing of real satellite HSI data.

1) *Sparse Unmixing:* The abundances are estimated using the Sparse SU algorithm [22] with endmembers from UAV platform.
2) *Endmember Variability:* The abundances are estimated using the ELMM algorithm [16] with endmembers from UAV platform.
3) *Mismodeling Effects:* The abundances are estimated using the RUSAL algorithm [19] with endmembers from UAV platform.
4) *Deep Learning Method:* The abundances are estimated using the uDAS algorithm [30] with endmembers from UAV platform.

Since no reference abundance information is available for the real satellite HSI, superior performance of the proposed IIDSU can be noticed and shown in Tables II and III and Figs. 4–6.

As can be seen in Tables II and III, for all compared unmixed models, the unmixing methods in the IIDSU framework significantly outperform the original unmixing methods for all datasets, with the best results underlined in bold italics. These results show that the use of IID-embedded SU models can significantly improve the unmixing performance by reducing the spectral variability compared to those using the original satellite HSIs as input. In particular, the proposed IIDSU algorithm achieves robust unmixing performance compared

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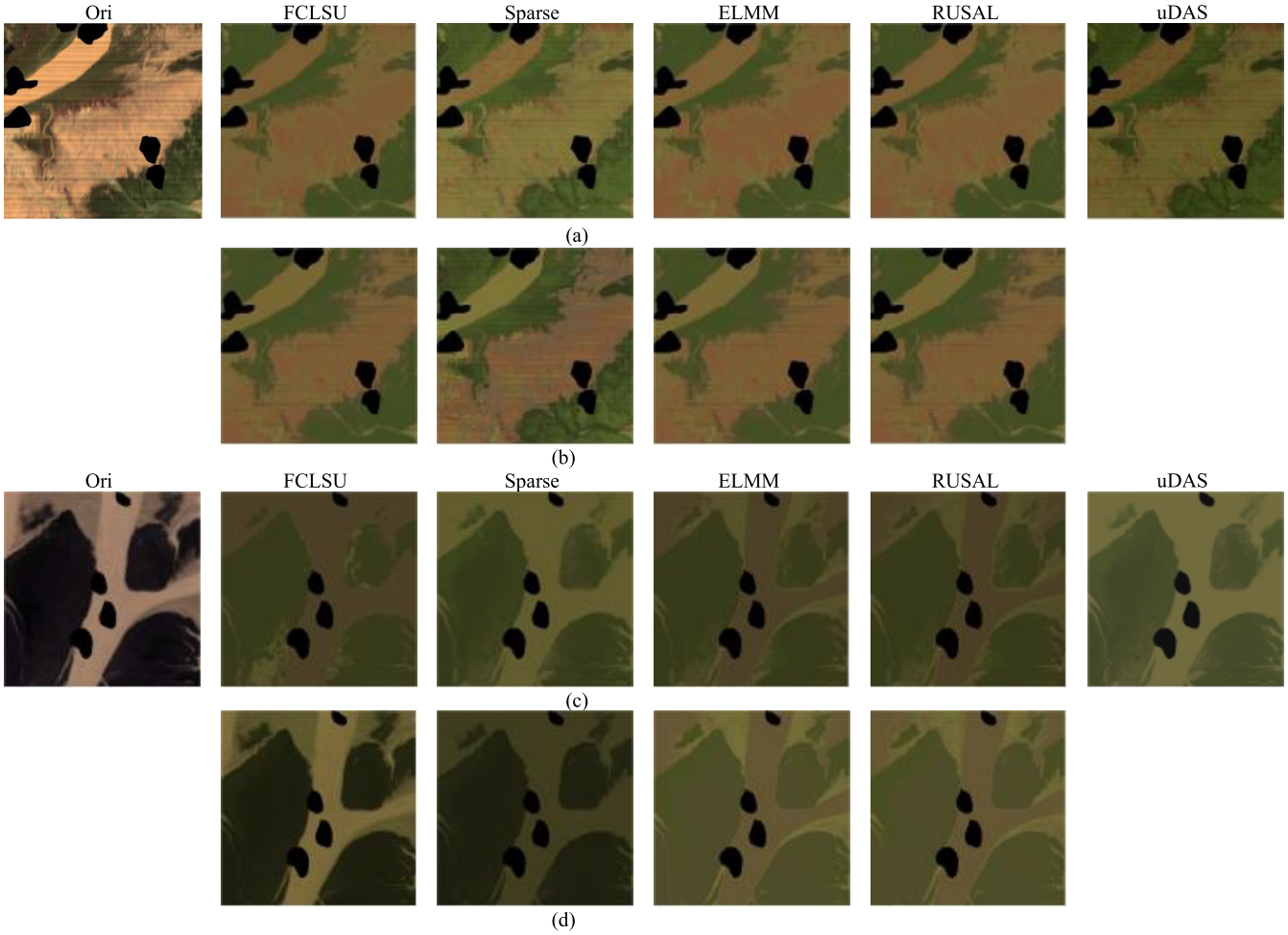


Fig. 5. Reconstructed results of unmixing (YRD_ZY102D image) obtained by the original model and the proposed IIDSU model using FCLSU, Sparse SU, ELMM, RUSAL, and uDAS methods for two subblocks. (a) Subblock1-Ori. (b) Subblock1-IIDSU. (c) Subblock2-Ori. (d) Subblock2-IIDSU.

to traditional SU algorithms. The experimental results are consistent with the physical model, that is, the effectiveness and robustness of the model proposed in this article in solving the problem of the existence of spectral variability in real data is verified. Together, that means that in future work, extending the unmixing model to the IIDSU framework can achieve more effective results, regardless of whether the model is modeled for spectral variability or not.

SUBBLOCK1: For the YRD_Orbit dataset, we can see that the improvement obtained by IIDSU is a reduction of about 0.01 in RMSE and an increase of 0.3–0.8 in PSNR. For the YRD_ZY102D dataset, IIDSU demonstrates its superiority with a reduction in RMSE of 0.02–0.04 and an increase in PSNR of 1–3.7. And for the YRD_GF5 dataset, the advantages are as high as 0.04–0.06 for error reduction and almost 3 for PSNR increase.

SUBBLOCK2: For the YRD_Orbit dataset, we can see that the improvement obtained by IIDSU is RMSE reduction of 0.03–0.04 and a PSNR increase of up to 7.4. For the YRD_ZY102D dataset, IIDSU reveals its superb performance with a reduction in RMSE of 0.04–0.06 and an increase in PSNR of 1.7–2.4. And for the YRD_GF5 dataset, there is a reduction in RMSE of 0.03–0.04 and an increase in PSNR of 0.07–2.02.

As shown in Tables II and III, FCLSU gives worse results with higher RMSE in the real data due to the spectral variability among the physics-based methods. Sparse SU gives better results than FCLSU due to the sparsity constraint on the abundance component. Considering the ELMM model aims at acquire the scale factors as well as the RUSAL model aims at gain the error term, the ELMM model and the RUSAL model have similar results on the basis that they both used to model spectral variability. Notably, the data-driven deep learning method uDAS also imposes sparse constraints, but does not yield significant benefits, and the reconstruction errors are larger than the sparse unmixing reconstruction results, possibly due to limited learning capacity due to the complexity of the real-world datasets.

Since sparse SU [22], ELMM [16], RUSAL [19] as well as the deep learning method [30] represent different perspectives on the data modeling, resulting in different levels of effectiveness on different datasets, but within the framework of the model proposed in this article, all physics-based comparison methods can achieve significantly superior results.

1) Analysis of the Impact of Different Satellite Sensors: For subblock1, the YRD_Orbit dataset has the smallest error, while the YRD_ZY dataset has an intermediate error and the YRD_GF5 dataset has the largest error; for subblock2, the YRD_Orbit dataset has the smallest error, while the YRD_ZY and YRD_GF5 datasets have a larger error. Theoretically, the YRD_GF5 and YRD_ZY datasets should perform better due to the higher number of spectral bands. The worse results can

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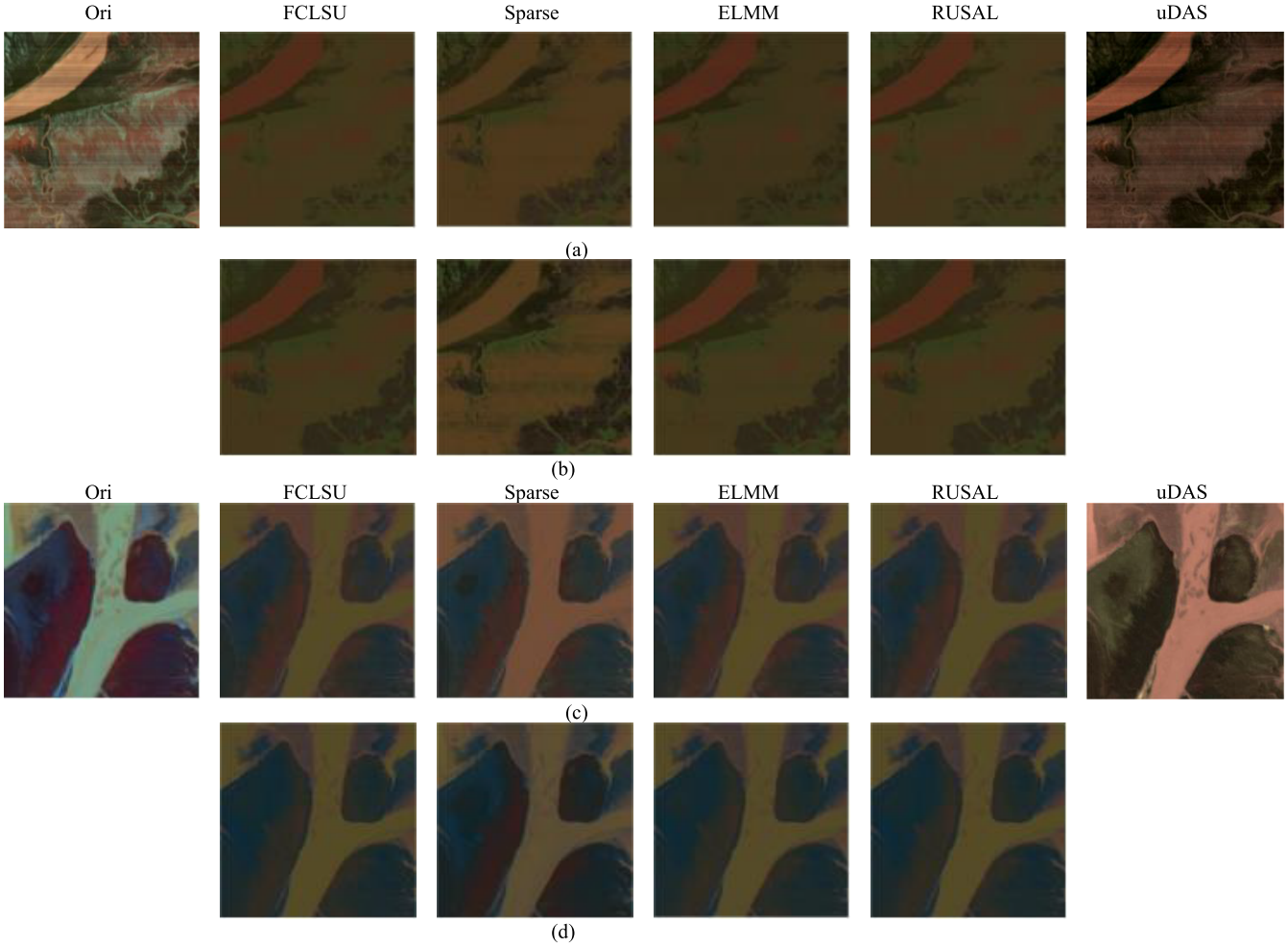


Fig. 6. Reconstructed results of unmixing (YRD_GF5 image) obtained by the original model and the proposed IIDSU model using FCLSU, Sparse SU, ELMM, RUSAL, and uDAS methods for two subblocks. (a) Subblock1-Ori. (b) Subblock1-IIDSU. (c) Subblock2-Ori. (d) Subblock2-IIDSU.

be attributed to the fact that, despite the maximum preservation of spectral information during the fusion process, unavoidable spectral distortion can still have an impact. Another reason for the poorer performance of YRD_GF5 may be due to the large difference between the data acquisition time and the endmember acquisition time.

2) Analysis of the Impact of Different Subblocks: Subblock1 is a more complex selection of species of interest and contains all the species of the UAV spectral library acquired from the UAV data, while subblock2 is selected from the inlet area and contains fewer species than subblock1. From the comparison of Tables II and III, it can be concluded that the results of the sparse SU of subblock2 show a consistent superiority among the different datasets, which could account for the higher clustering of feature types and the more significant sparsity characteristics. Because of the complex ground distribution of satellite HSIs of large scene in real data, there are often many unknown categories in addition to the six selected feature types of the UAV spectral library, which leads to a significant unmixing error.

3) Analysis of the Impact of Different Parameter Settings: Fig. 7 shows the parameter sensitivity analysis of the proposed IIDSU in terms of two regularization parameters σ and γ in (26) in the YRD_Orbit data of subblock1. The range of values for the parameters is essentially taken from the literature [16], [18], [22], [25]. It can be seen from Fig. 7 that the value of σ has only a small effect on the RMSE while the regularization parameter γ for the sparsity-inducing terms has an optimal range of values.

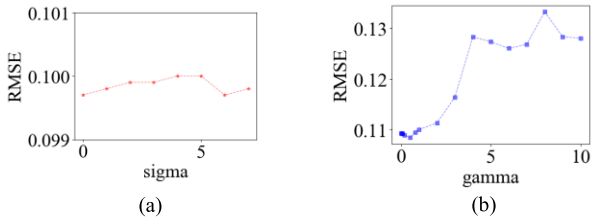


Fig. 7. Parameter sensitivity analysis of the proposed IIDSU in the YRD_Orbit data of subblock1. (a) σ and (b) γ.

4) Analysis of the Computational Time: Tables IV and V show the computational time of different SU methods without using the IIDSU framework and with using the IIDSU framework for three datasets of two subblocks. The FCLSU without using the ADMM implementation has a shorter computational time as no iterative operations are performed. Besides, considering the complexity of the real experimental datasets used in this article, the modeling of RUSAL as well as ELMM models always take a longer computational time,

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TABLE III
RESULTS OF THE THREE DATASETS OF SUBBLOCK2

Subblock2	Methods	RMSE		PSNR	
		Ori	IID	Ori	IID
YRD_Orbit	FCLSU	0.1438	*0.1147*	18.1442	*19.7591*
	Sparse	0.1458	***0.1098***	16.7564	***24.1462***
	ELMM	0.1429	*0.1109*	17.9828	*20.3508*
	RUSAL	0.1424	*0.1105*	17.9552	*20.1935*
	uDAS	0.1491		24.12	
	Ideal	0		∞	
		Ori	IID	Ori	IID
YRD_ZY102D	FCLSU	0.2067	*0.1603*	10.5058	*12.1697*
	Sparse	0.1562	***0.1173***	12.8086	***15.2026***
	ELMM	0.2049	*0.1586*	10.5713	*12.2712*
	RUSAL	0.2057	*0.1592*	10.5238	*12.2232*
	uDAS	0.3147		10.95	
	Ideal	0		∞	
		Ori	IID	Ori	IID
YRD_GF5	FCLSU	0.2083	*0.1755*	14.1961	*17.1002*
	Sparse	0.2011	***0.1617***	16.6395	***18.6622***
	ELMM	0.2042	*0.1732*	14.4258	*17.2898*
	RUSAL	0.2061	*0.1675*	18.5300	*18.6071*
	uDAS	0.2075		18.10	
	Ideal	0		∞	

TABLE IV
COMPUTING TIME (IN SECONDS) OF DIFFERENT SU METHODS FOR THE THREE DATASETS OF SUBBLOCK1

METHODS	YRD_Orbit	YRD_ZY102D	YRD_GF5
FCLSU	43.73	106.97	69.12
IIDSU	48.03	90.29	69.91
Sparse	683.77	2.98e+3	2.35e+3
IIDSU	614.35	2.20e+3	1.83 e+3
ELMM	1.66e+3	7.52e+3	1.69e+4
IIDSU	1.68e+3	5.73e+3	1.11e+4
RUSAL	1.22e+3	4.61e+3	2.15e+3
IIDSU	230.76	4.82e+3	1.88e+3
UDAS	681.13	2.77e+3	4.01e+3

TABLE V
COMPUTING TIME (IN SECONDS) OF DIFFERENT SU METHODS FOR THE THREE DATASETS OF SUBBLOCK2

METHODS	YRD_Orbit	YRD_ZY102D	YRD_GF5
FCLSU	31.80	45.50	52.68
IIDSU	35.60	51.28	53.11
Sparse	468.56	852.94	1.05e+3
IIDSU	468.81	848.10	1.01e+3
ELMM	1.11e+3	4.89e+3	1.30e+4
IIDSU	1.15e+3	5.27e+3	1.06e+3
RUSAL	463.61	1.26e+3	823.73
IIDSU	218.93	798.18	824.29
UDAS	595.02	1.52e+3	2.83e+3

where ELMM essentially having the longest computation time. Most importantly, the computational time of the SU method using the IIDSU framework has been optimized in comparison to the original SU model.

V. CONCLUSION

In this article, a novel UAV-to-satellite collaborative hyperspectral remote sensing image unmixing model based on IID is proposed for hyperspectral unmixing. The introduced IID can significantly reduce the spectral variability that occurs in the imaging process of HSIs. In addition, by integrating two different scale data sources of UAV HSI and satellite HSI, the characteristics of UAV, namely, high resolution can guarantee the accuracy of target endmember extraction, are fully exploited. To be more specific, the proposed method consistently achieves more robust and effective results among the comparative unmixing methods for modeling spectral variability. By embedding the IID model into the novel UAV-to-satellite collaborative LMM model for matrix modeling, the separated abundance matrix can be obtained while reducing the spectral variability by using a prior UAV spectral library. Experimental results on three different hyperspectral datasets show that the proposed method has excellent performance and the efficiency of the method is further demonstrated. However, the IIDSU framework has certain requirements on the data, i.e., the phenological stage of acquisition of the UAV HSI and the satellite HSI need to be similar. In addition, to further validate the results of SU, the use of hyperspectral data with higher spatial resolution or field exploration could be considered in the future.

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